



Sem. V Unit 2

Decision Making & Problem Solving



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Sem V Unit 2 Decision Making & problem Solving

- 1. Cognitive bias- types, identifying bias, case studies; Techniques to reduce cognitive bias & case studies**
2. Decision-making models- types; decision-making in teams- tools & matrices (SWOT, decision-making tree)
3. Organisational Climate & Culture
4. Understanding change management
5. Resistance to change, empowerment & management strategies

HOW TO APPROACH THIS COURSE



- All **content slides** will be shared.



- Make your **own notes** in class.



- **Conceptual understanding** and **appropriate reasoning** are important from an examination point of view.



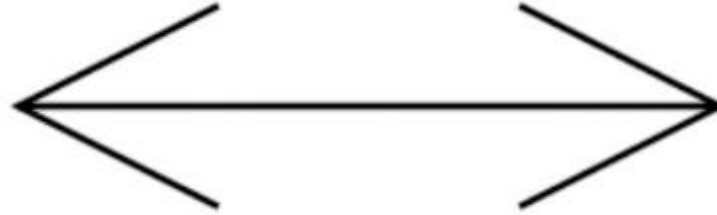
- But **DO NOT STUDY ONLY FOR MARKS.**



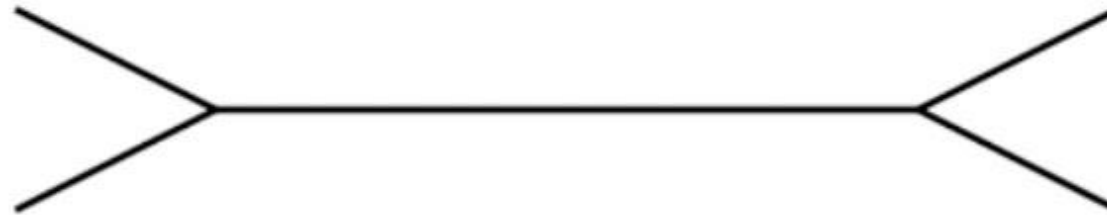
**THIS COURSE IS VERY IMPORTANT AND
USEFUL FOR YOUR FUTURE PROFESSIONAL LIFE.**







Optical Illusion







Cognitive Illusion

Bat and ball together cost Rs 110

Bat costs Rs 100 more than the ball

What are their individual costs?





One Sunday there were 20 lotuses in a pond.
The next day the lotuses were doubled to 40.
On the third day, it became 80 lotuses.

The pond was full of lotus by **20** days.

On which day had $\frac{1}{2}$ th of the entire pond been filled with lotus?



HOW MANY HOLES DOES THIS T-SHIRT HAVE?



Marc looks at Anne, and Anne looks at Simon.

Marc is married, and Simon is single.

Does a married person look at an unmarried person?



KBC



Catch the ball

$$2 + 2 =$$

$$24 \times 17 = ?$$



THE NEW YORK TIMES BESTSELLER

THINKING, FAST AND SLOW



DANIEL
KAHNEMAN

WINNER OF THE NOBEL PRIZE IN ECONOMICS

"[A] masterpiece . . . This is one of the greatest and most engaging collections of insights into the human mind I have read." —WILLIAM EASTERLY, *Financial Times*



Kahneman & Tversky

JPIP

JPIP



JPIP

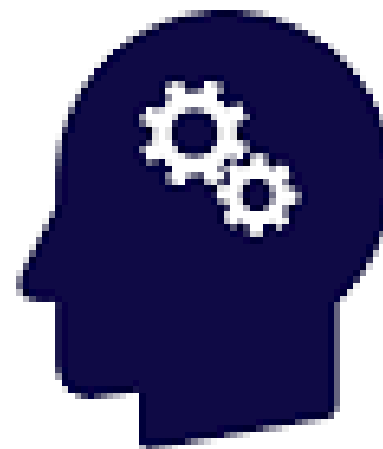
system 1



fast
unconscious
automatic
everyday decisions

JPIP

system 2



slow
conscious
effortful
complex decisions

TWO SYSTEMS OF THINKING

Stanovich & Kahneman

SYSTEM 1

FAST, AUTOMATIC, UNCONSCIOUS



DANIEL KAHNEMAN

Nobel Laureate in
Economics

KEY FEATURES

- ⌚ Very fast
- 🧠 Automatic
- 👁️ Unconscious
- ⚡ Low effort
- 🧩 Heuristic-based
- 🎯 Prone to biases



AUTOMATIC AL

The Hare
"Jumping to conclusions"

WHAT IT DOES

- Recognizes patterns
- Fills in the gaps
- Makes quick judgments
- Works immediately

EXAMPLES

- ✓ Instincts
- ♥ Gut feelings
- 🔥 Habits
- 💡 First impressions

"System 1 is like the hare:
fast, intuitive, and always in a hurry."

SYSTEM 2

SLOW, EFFORTFUL, CONSCIOUS

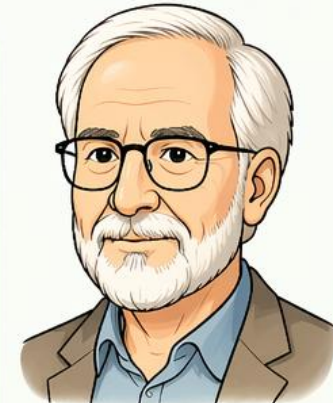


REFLECTIVE ROY

The Tortoise
"Thinking it through"

KEY FEATURES

- ⌚ Slower
- ⚙️ Effortful
- 🔍 Conscious
- 🔋 High effort
- 📋 Rule-based
- 🛡️ More reliable



KEITH STANOVICH

Cognitive Scientist
**SYSTEM 2 =
REFLECTIVE &
ALGORITHMIC**

WHAT IT DOES

- Plans ahead
- Solves complex problems
- Inhibits impulses
- Applies rules and logic

EXAMPLES

- 📊 Math problems
- 🧩 Logical reasoning
- 📅 Planning
- 🛡️ Self-control

"System 2 is like the tortoise:
slow, steady, and gets things right."

THE RELATIONSHIP

System 1 does the judging.



System 2 does the reasoning.



Together, they shape every decision.





Examples



Very Very Important

- The brain is not **literally** divided into two systems
- System 1 and 2 work in **tandem**, not as separate entities
- **Both** systems can be biased and can make mistakes - neither one is categorically “good” or “bad”



Intuitive guess at the exam?
The first answer that comes to mind?
Read the question carefully.

Mindful practice

Being in the flow



Practice properly to convert system 2 into Auto mode.

System 2: Interview

Take a few seconds' **pause** to recollect



Our brain is highly optimized for certain things
& **Not** optimized for other things

This is how it is!!!

Quiz

1 You ask an AI:

“Which of these two bridge designs is safer?”

AI gives a confident, technically impressive answer in 10 seconds.

Which response demonstrates System 2 thinking?

- A. “AI is probably correct—it sounds convincing.”
- B. “I'll use the answer because AI knows engineering.”
- C. “I'll identify the assumptions, check the calculations and compare the evidence.”
- D. “I'll ask AI the same question again.”



**2 An experienced mechanic hears a machine for 2 seconds and says:
“Misalignment.”**

What is most likely happening?

- A. Random guessing
- B. System 1 pattern recognition
- C. System 2 calculation
- D. Confirmation bias



3 AI tells an engineer:

“Based on the symptoms, the most likely cause is bearing failure.

Confidence: 94%.”

What should the engineer do next?

- A. Accept it because AI says 94%
- B. Ask AI to explain it again
- C. Check alternative causes and supporting evidence
- D. Reject AI automatically



Groups of 5

- Discuss and Present examples of each of systems 1 and 2
- Personal as well as professional (engineering) examples



System 1

- Picking up ph
- Biometric check

system 2

- Journal
- Cooking for novice
- planning for vacation
- Problem in presentation: check



Cognitive Bias

Heuristics



A rule, strategy or mental shortcut that one can use to derive a solution to a problem

Rule of thumb

Cognitive Bias



Bias: Inclination/tendency



Normal human brain function: bounded rationality



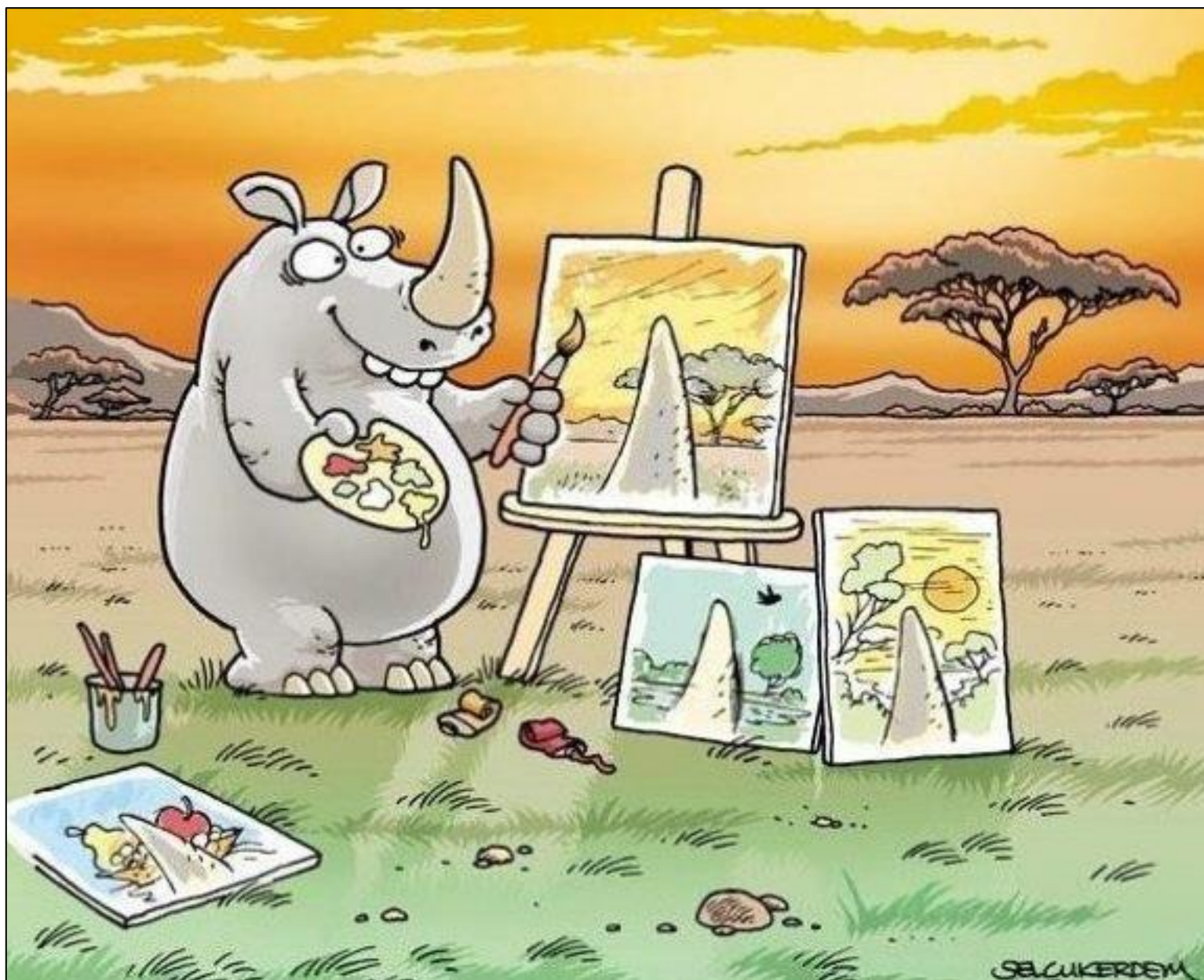
Gap between how we ought to reason ideally & how our brain actually reasons



We can not avoid biases, but we can manage them

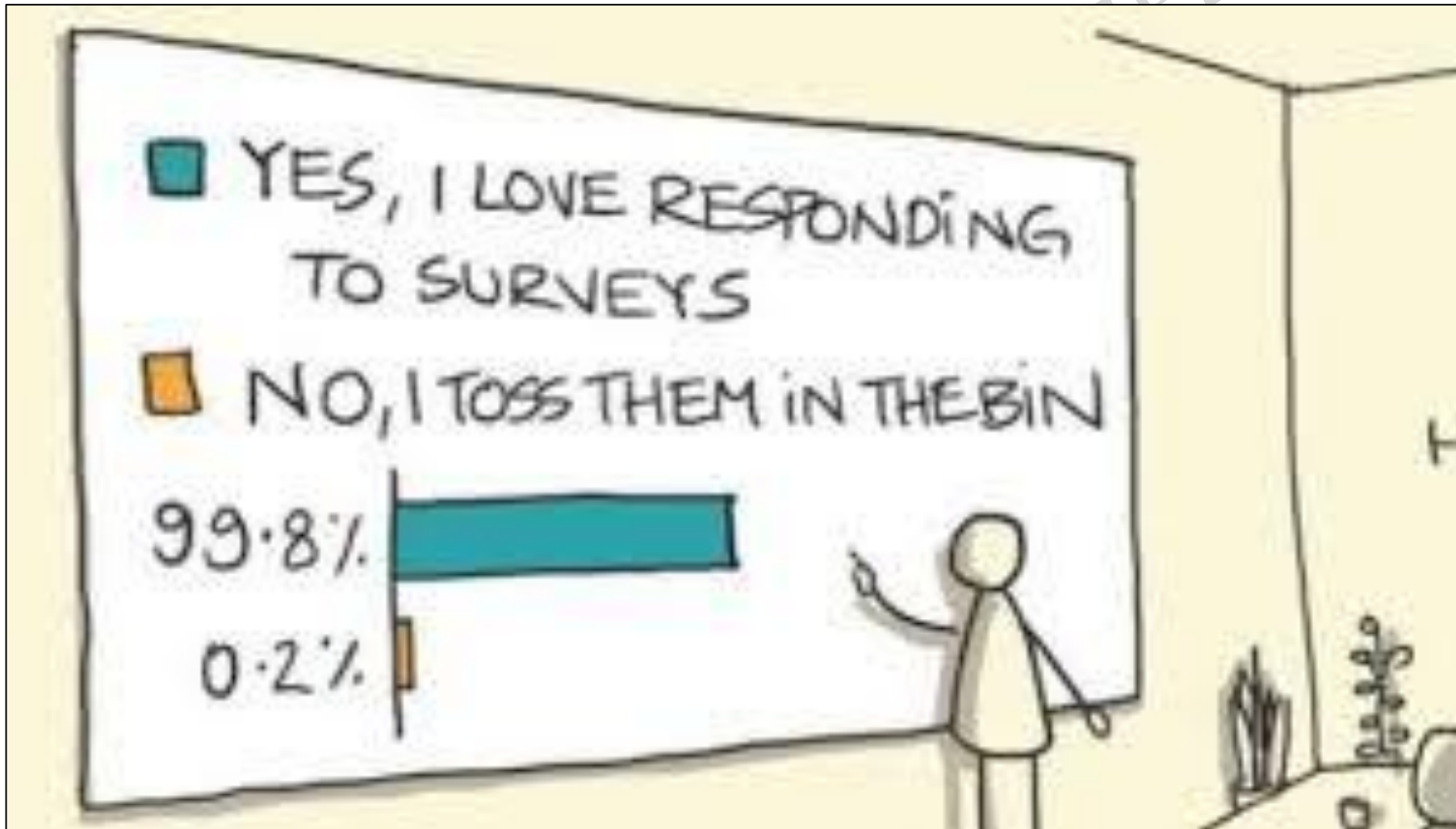


More than **100** documented biases



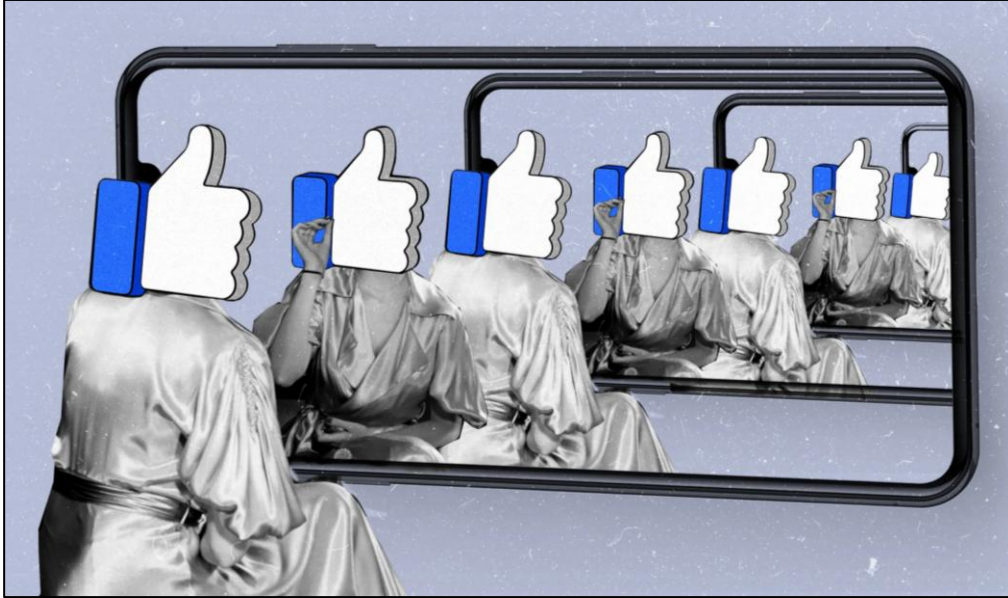
Scientists may selectively report results that support their hypothesis while **overlooking or downplaying** conflicting evidence.





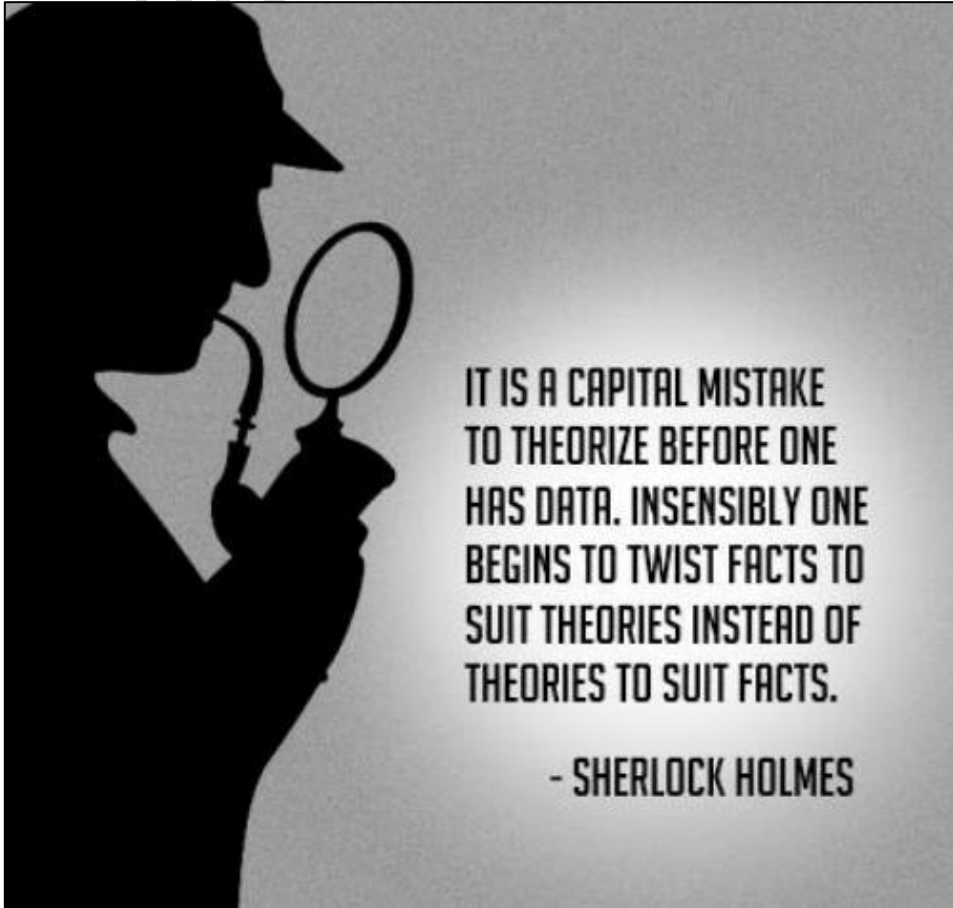


1 Confirmation Bias

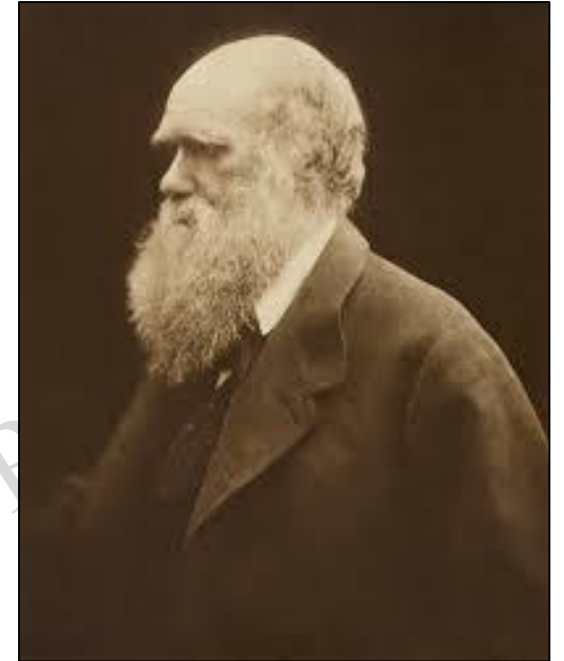


We love to interact with people who agree with us, which adds to confirmation bias.

Application: With whom I should discuss this?
Choose colleagues with different perspectives.



In writing answers
Thinking.....



Sir Charles Darwin



Confirmation Bias in Engineering — 3 Applications

- 1. Fault Diagnosis:** An engineer suspects a bearing failure and notices evidence supporting it while ignoring signs of electrical or alignment problems.
- 2. Design Decisions:** An engineer prefers their own design and selectively focuses on test results that support it while dismissing contradictory results.
- 3. Project Management:** A team believes a project will succeed and interprets delays or warning signs as “temporary,” rather than reconsidering the original plan.



What to do?

- Seek out **contradictory** information and constructive criticism.
- Ask **apt questions** while gathering information.
- **Decide accordingly**



5 PRACTICAL STEPS OF DEBIASING



Use these steps for *any* decision or problem.



Based on: 1) S. L. Rivers & D. L. Schuman (2006) – How to Think Straight About Psychology

2) Gerd Gigerenzer (2013) – “Why Heuristics Work” 3) Daniel Kahneman (2011) – Thinking, Fast and Slow

EXAMPLE: CONFIRMATION BIAS

(We look for evidence that supports what we already believe and ignore the rest.)

1



PAUSE & NOTICE

Pause before deciding. Notice your immediate thoughts, feelings and assumptions.

“What am I assuming right now?”

2



SEEK DISCONFIRMING EVIDENCE

Actively look for information that goes against your belief or hypothesis.

“What evidence would prove me wrong?”

3



CONSIDER ALTERNATIVES

List other possible explanations or options. Compare them fairly.

“What else could be causing this?”

4



EVALUATE OBJECTIVELY

Use data, facts, or experiments. Rate each option based on evidence, not intuition.

“What does the data actually show?”

5



DECIDE & REVIEW

Make the decision. Later, review the outcome and learn. Update your thinking.

“What did I learn? How will I improve next time?”



Situation: A motor in a production line is not working.

Engineer's first thought: “The bearing must be faulty.”



Engineer actively looks for evidence **against** his assumption: checks electrical supply, sensor readings, alignment, overload, etc.



Possible causes listed:

- | | |
|--------------------|---------------------|
| 1) Bearing failure | 2) Electrical issue |
| 3) Misalignment | 4) Overload |



Collects data/observations and compares:

- | | |
|----------------------|-----------------------------|
| • Vibration normal | • Power fluctuation present |
| • Temperature normal | • Recent overload alarm |

Electrical issue has stronger evidence.



Fixes the electrical issue → Motor works.

Reviews the process and updates checklist to avoid similar bias in future.



GOOD ENGINEERS USE SYSTEM 2 THINKING:

Slow down, question, test, and then decide.



KEY TAKEAWAY: Being right is less important than being accurately informed.



1



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OPERATION SINDOOR



Laila Ki Ungliyan

Recipe by



Total Time:
15-30
minutes



Serves: 2

Nutrition facts:

240 calories, 9 grams fat



The Framing Effect

A



**Old
Furniture**

B



**Vintage/ Rustic
Furniture**



Smartiac.org



What is healthier?

99% fat-free food

Or

food containing 1% fat



What Does a
Fat-Free Label
Really Mean?



Video

- [Am I audible?](#)

A college placement cell gives students two descriptions of the **same company**:

Frame A — Negative “20% of employees leave this company within the first year.”	Frame B — Positive “80% of employees stay with this company for at least one year.”
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Where is the framing bias?

The **information is the same**, but the **presentation changes the emotional response and judgment**.

Debiasing

1. **Recognize:** “Am I reacting to the wording rather than the actual information?”
2. **Reframe:** Convert negative → positive, or positive → negative.
20% leave = 80% stay
3. **Check:** Ask: “*What are the facts?*”
Also consider salary, growth, work culture, location, job security, etc.
4. **Decide:** Make the decision based on the **complete evidence**, not on which frame sounds better.

“Same facts + different presentation = different reality.”



Application: debiasing framing bias

- Understand the framing of information and its consequences for us
- Frame your answer in an interview, etc.

I failed this mock test. I discovered areas for improvement."

Anchoring



Limit: **12** per person

7 average purchase



No limit

3 average purchase

Anchoring bias binds you to your first impression



Various anchors



Here, the word anchor is used when we compare various scenarios.

- **Life anchors:** Friends, colleagues, initial/previous relation
- **Buying** something: initial price
- **Rating something:** Previous incidence, etc





Anchoring Bias – Debiating Example

Example:

A laptop is initially listed at ₹80,000 and then offered at **₹55,000**. You feel ₹55,000 is a great deal because ₹80,000 became your **anchor**.

Debias: Before judging the price, independently estimate the laptop's reasonable value and compare **multiple alternatives/prices**.

Application

- 1. Shopping & Negotiation:** The first price mentioned influences what seems “cheap” or “expensive.” → **Debias:** Determine your own reasonable price before seeing the offer.
- 2. Performance Evaluation:** An initial impression or rating influences later judgments.
→ **Debias:** Evaluate using objective criteria and fresh evidence.
- 3. Decision Making:** The first idea or suggestion becomes the reference point for subsequent choices. → **Debias:** Generate alternatives **before** evaluating the first suggestion.



You went for a movie
At interval you feel that it is boring

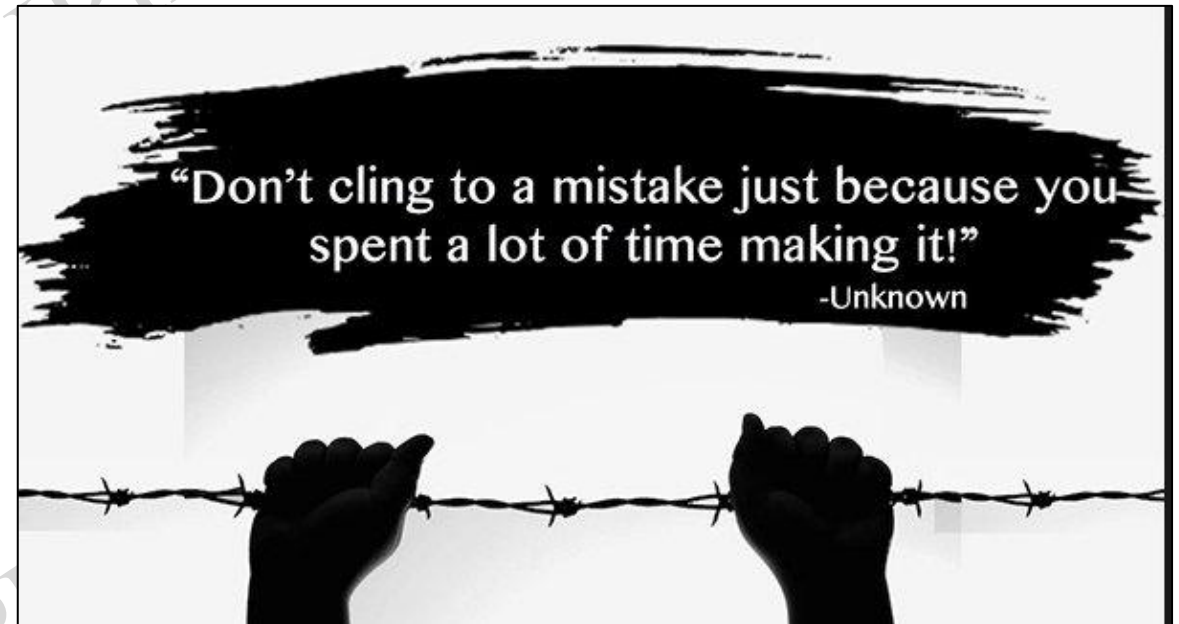
What will you do?



The Great Battle Maratha & Mughals 1680 To 1707

Sunk cost fallacy

Commitment bias: We continue to support our past decisions despite new evidence suggesting that it isn't the best course of action



Sunk cost fallacy



“I might as well keep going to a bad/useless class that I paid for.”

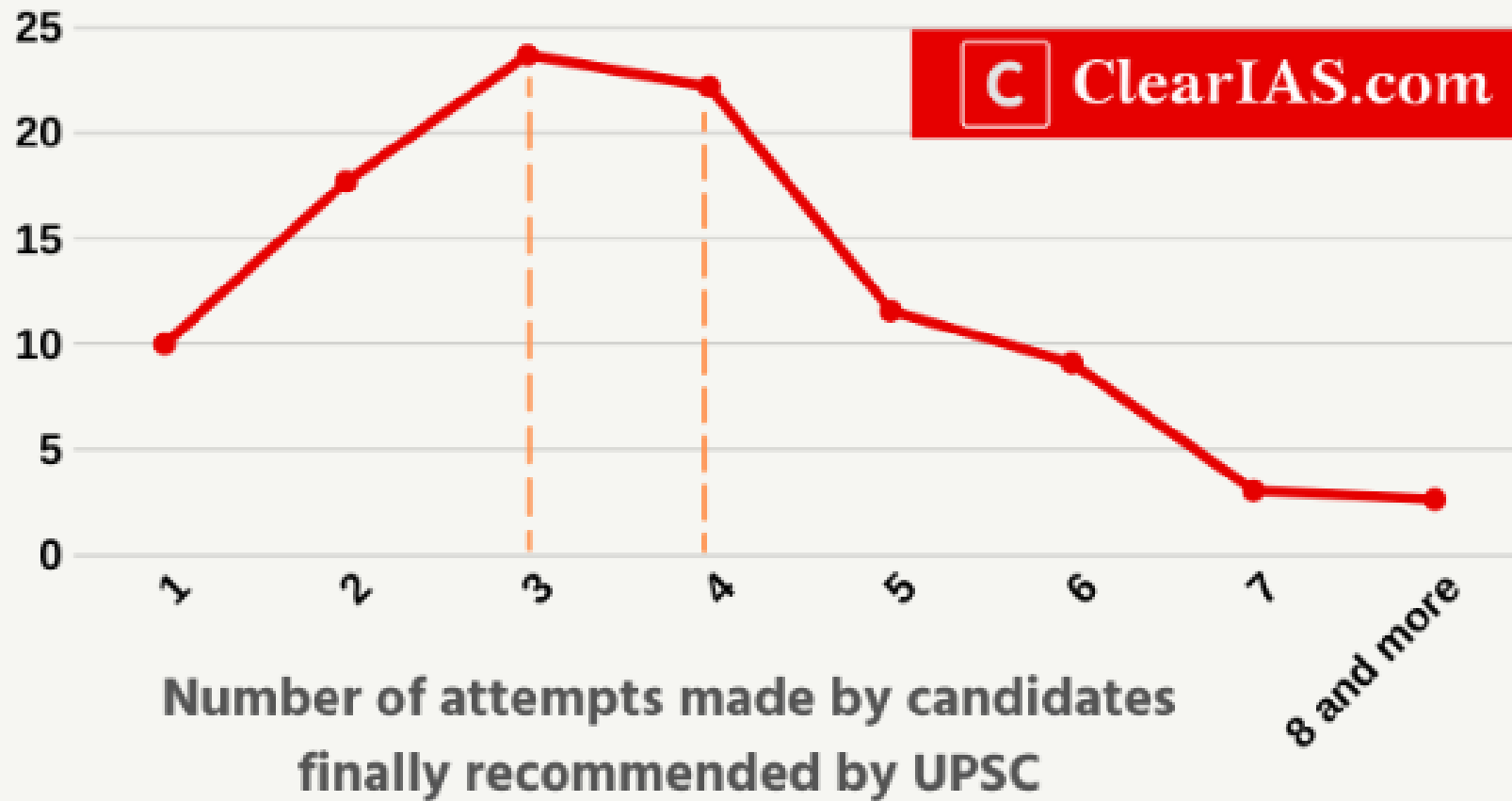


“I might as well continue dating someone bad for me because I’ve already invested so much in them.”



“I might as well keep eating because I already bought the food.”

Percentage of candidates







Sunk Cost Fallacy

Example:

You have already spent ₹50,000 on a course, but after attending it for a month, you realize it is not useful.

You continue **only because you have already paid.**

Debias:

Ask: **“If I had not spent anything so far, would I choose to continue today?”**

Focus on **future costs and benefits**, not past investment.



Debiasing sunk cost fallacy

1.Engineering Projects: Continuing a failing project because “we have already invested ₹10 crore.” → **Debias:** Evaluate only the future cost, benefits and feasibility.

2.Technology/Equipment: Continuing to repair an old machine because “we have already spent so much on it.” → **Debias:** Compare future repair costs with the cost and benefits of replacement.

3.Personal/Business Decisions: Continuing a job, course, product or business venture because of time/money already invested. → **Debias:** Ask, “**What would I choose if I were starting today?**”

Availability bias



Estimating what is more likely by what is more **available** in memory,

which is biased towards
vivid, unusual, or emotionally charged examples

Caught In The Lift



9/11



26/11



Availability Bias – Engineering Example

Example:

An engineer recently witnessed a **battery overheating incident** in the lab. When evaluating a new battery design, the engineer immediately believes overheating is the **most likely failure**, even though historical data shows that electrical connection failure is much more common.

Debiasing:

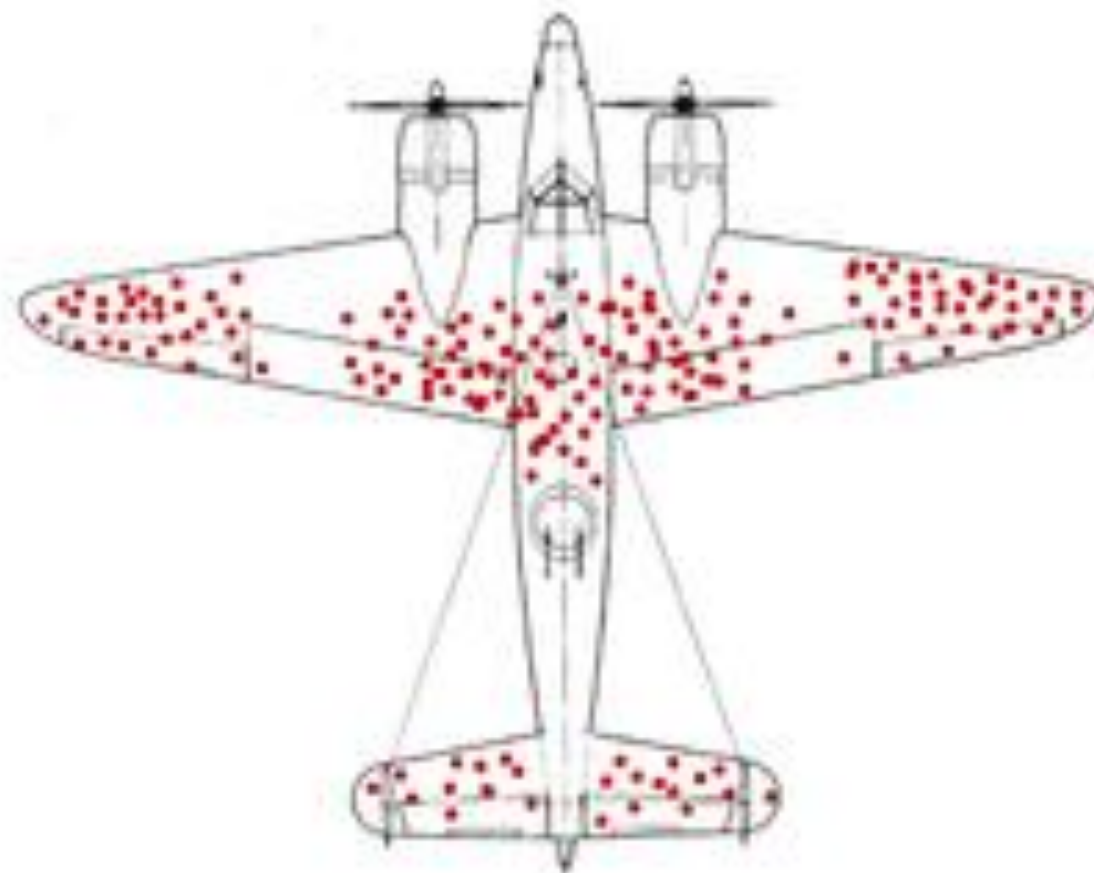
Ask: “**Am I judging the probability based on what I remember easily, or on actual failure data?**” → Check **base rates, historical records and multiple failure modes** before deciding.



Application

- Am I focusing on available information in my memory only? Or can I stretch it?
- Students might overestimate the importance of the topics they've recently studied or those frequently appearing in mock tests while **underestimating less common but still critical topics**.
- **Medical Decisions:** A recently heard story about a rare disease may make someone overestimate their own risk.
→ **Debias:** Consider prevalence and base rates.
- **Business/Investing:** A memorable success or failure may disproportionately influence an investment decision.
→ **Debias:** Use historical data and multiple sources rather than memorable examples.







6 Survivorship Bias

The tendency to concentrate on the people or things that "**survived**" some process and ignore those that didn't, or

arguing that a strategy is effective given the winners while **ignoring** a large number of losers.



Singing competition



Lottery winners



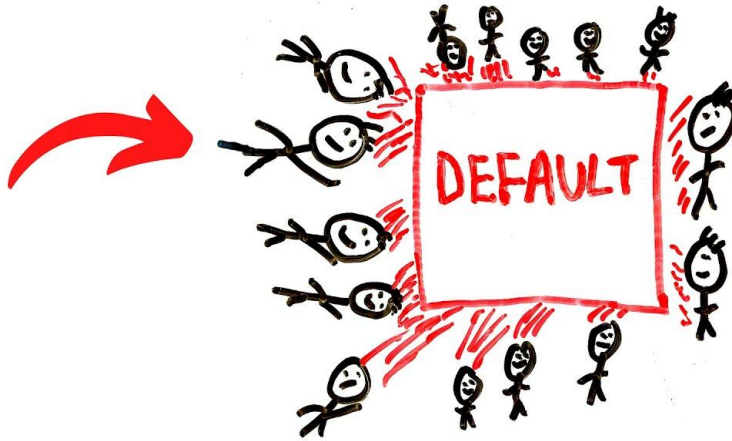
Chetan Bhagat
IIT Delhi



Application

- Avoid Blindly Copying Success Stories
- Customize the approach

STATUS QUO BIAS



- A preference for the maintenance of one's existing state of affairs.
- The bias intersects with other non-rational cognitive processes such as [loss aversion](#)



Example:

An engineering team continues using an **old software/tool** simply because “we have always used it,” even though a newer system is faster, safer and more efficient.

Debiasing:

Ask: “**If we were choosing this system today from scratch, would we still select the current one?**” Compare the current option with alternatives using objective criteria.

3 Applications

- 1.Engineering Design:** Continuing with a traditional design despite better alternatives.
- 2.Work Processes:** Following inefficient procedures because employees are accustomed to them.
- 3.Technology:** Avoiding new technology because changing the existing system feels risky or inconvenient.



Status Quo / Anchoring bias advantages?

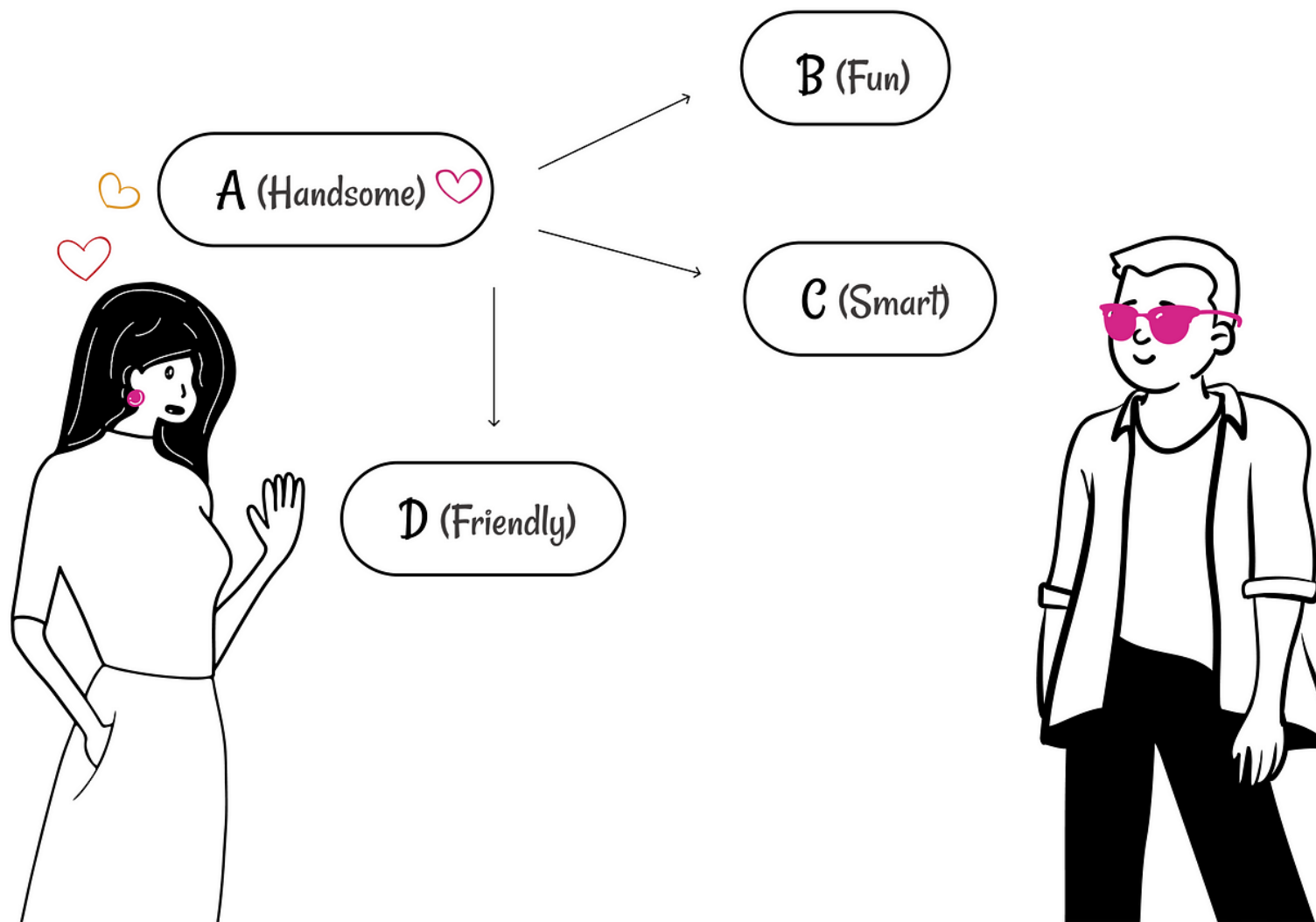


The Halo Effect

The "halo effect" is when one trait of a person or thing is used to make an *overall judgment* of that person or thing.

angroup.com

NN/g





Application

- Not getting stressed with outward impressions of others
- How can I manage my impression @interview?

Bias	Definition	Example	Debiasing
Halo Effect	One positive quality influences our overall judgment of a person.	A student speaks confidently, so the teacher assumes they are also highly competent.	Rate each quality separately using objective evidence.
Horn Effect	One negative quality influences our overall judgment of a person.	An employee is late once, so the manager assumes they are generally unreliable and incompetent.	Separate the behaviour from the person and examine evidence across different areas.

How to Debias Halo & Horn Effects

- 1 Separate the dimensions:** Evaluate each quality independently rather than letting one impression influence everything else.
- 2 Use evidence:** Base ratings on **specific behaviors, performance data and multiple observations**, not general impressions.
- 3. Delay the overall judgment:** Form the overall evaluation **only after** rating individual criteria separately.

Simple rule for students:

“Don’t let one bright spot—or one dark spot—colour the whole picture.”

BANDWAGON EFFECT



A psychological phenomenon where people do something because others are doing it.

Reels



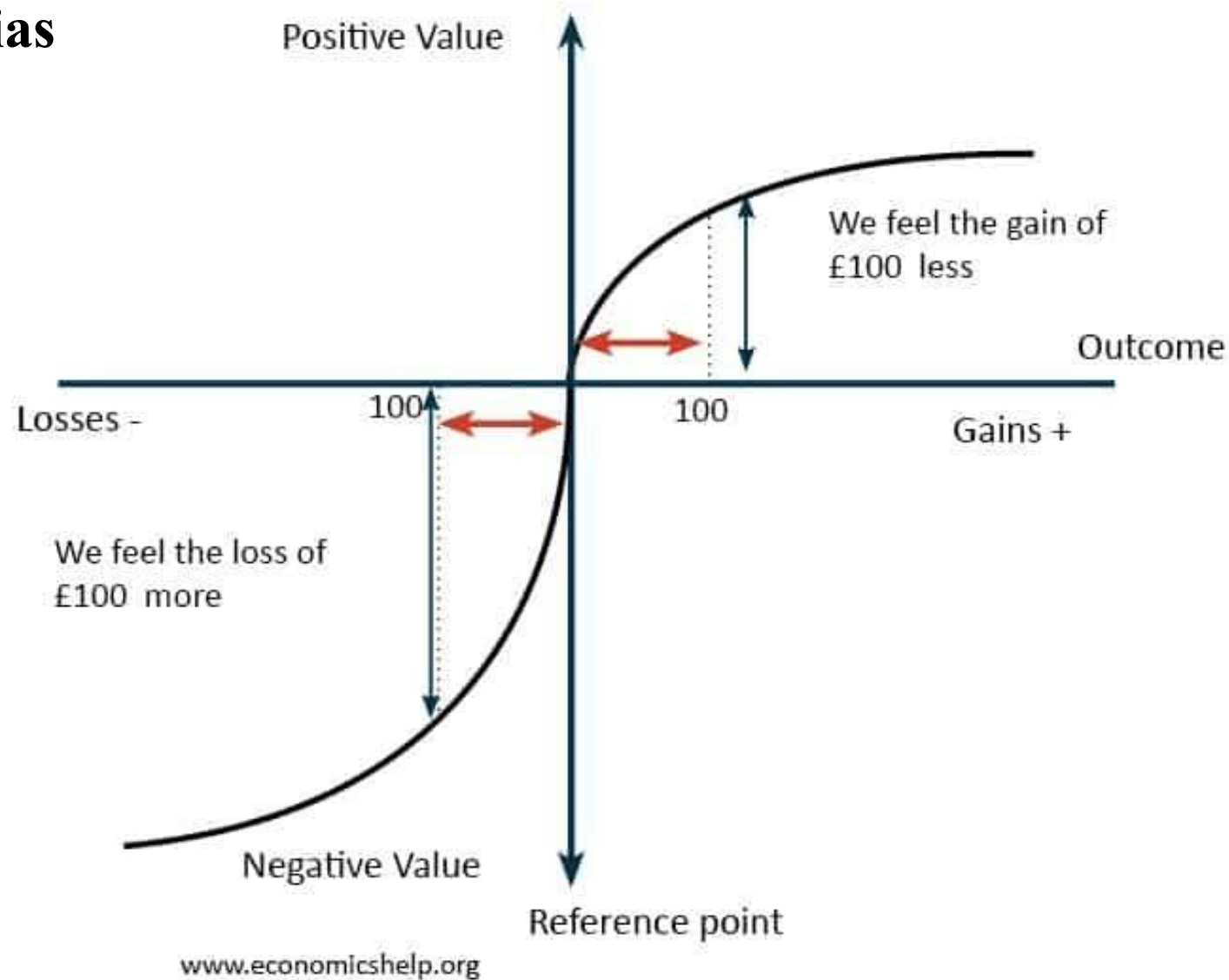
- Dancing [uncle](#)



Application

- Am I just a part of the herd? And following others?
- Ask reflect debias....

10 Negativity Bias





10 Negativity Bias

The tendency to give **more attention, weight and importance to negative information** than equally significant positive information.

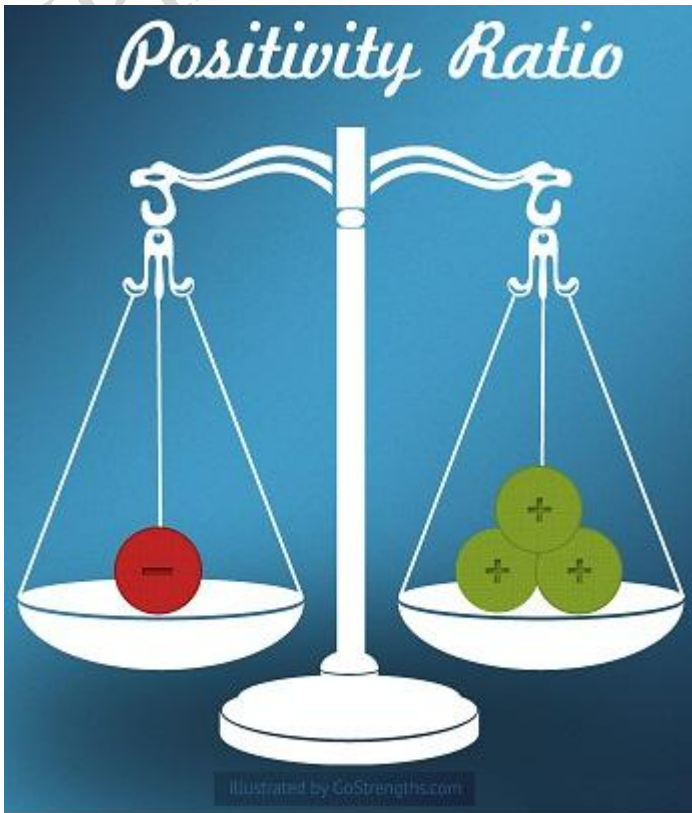
An engineer receives **9 positive comments and 1 critical comment** about a design, but keeps thinking about the one criticism and starts doubting the entire design.

Debiasing

- 1. Balance the evidence:** Deliberately identify both positive and negative information.
- 2. Check proportions:** Ask, *“Is this one negative event representative of the overall pattern?”*
- 3. Use objective data:** Base the decision on the **complete evidence**, not the most emotionally powerful information.

Simple rule:

“One negative signal is information—not the whole picture.”

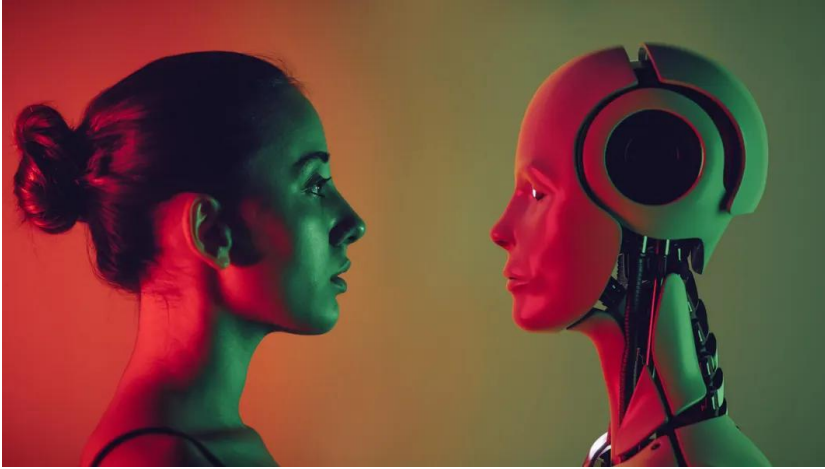


Source: Fredrickson, B. (2009). *Positivity*. Illustrated by GoStrengths.

We may need
triple positive strokes
to manage negative events.

Being objective about your performance

11 Automation Bias

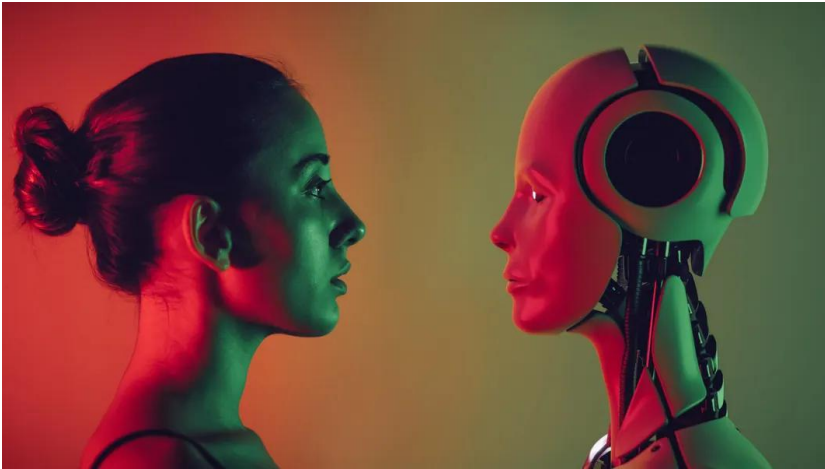


The tendency to over-rely on recommendations or decisions made by automated systems/AI, assuming they are correct.

Example:

An engineer accepts an AI diagnostic tool's suggestion that a machine has a bearing fault without independently checking other possible causes.

11 Automation Bias



Debiasing

Treat automation as advice, not truth: Ask, “**What evidence supports this recommendation?**”

Verify independently: Check data, assumptions and alternative explanations.

Know when to override: If the output conflicts with observations or domain knowledge, investigate before accepting it.

Simple rule:

“AI can assist the decision; it should not replace the decision-maker.”

1



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Case Study –

1. WHAT IS THE BIAS?

Identify the cognitive bias illustrated in the case.

2. WHY DID IT HAPPEN?

What thinking or information led the person/team toward this bias?

3. HOW CAN WE DEBIAS?

What could the person/team do differently to make a better decision?

Also provide 1 real life example of the same bias.



Task

- Spot your **own** biases
- Develop a debiasing strategy



In a nutshell,

- **A more mindful and strategic approach**
- **Self-awareness of biases**
- **Objective feedback**
- **Flexible strategies**
- **Balanced preparation and mindset**



Key points

*System 1 System2 Characteristics and examples

Biases

1. Confirmation Bias
2. Framing
3. Anchoring
4. Sunk Cost Fallacy
5. Availability
6. Survivorship bias
7. Status Quo Bias
8. Halo / Horn effect
9. Bandwagon effect
10. Negativity Bias
11. Automation Bias

Identify
Justify
Debias
Application
Etc.



Reference Books

